Acceleration Measurement Using Smartphones

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Aim

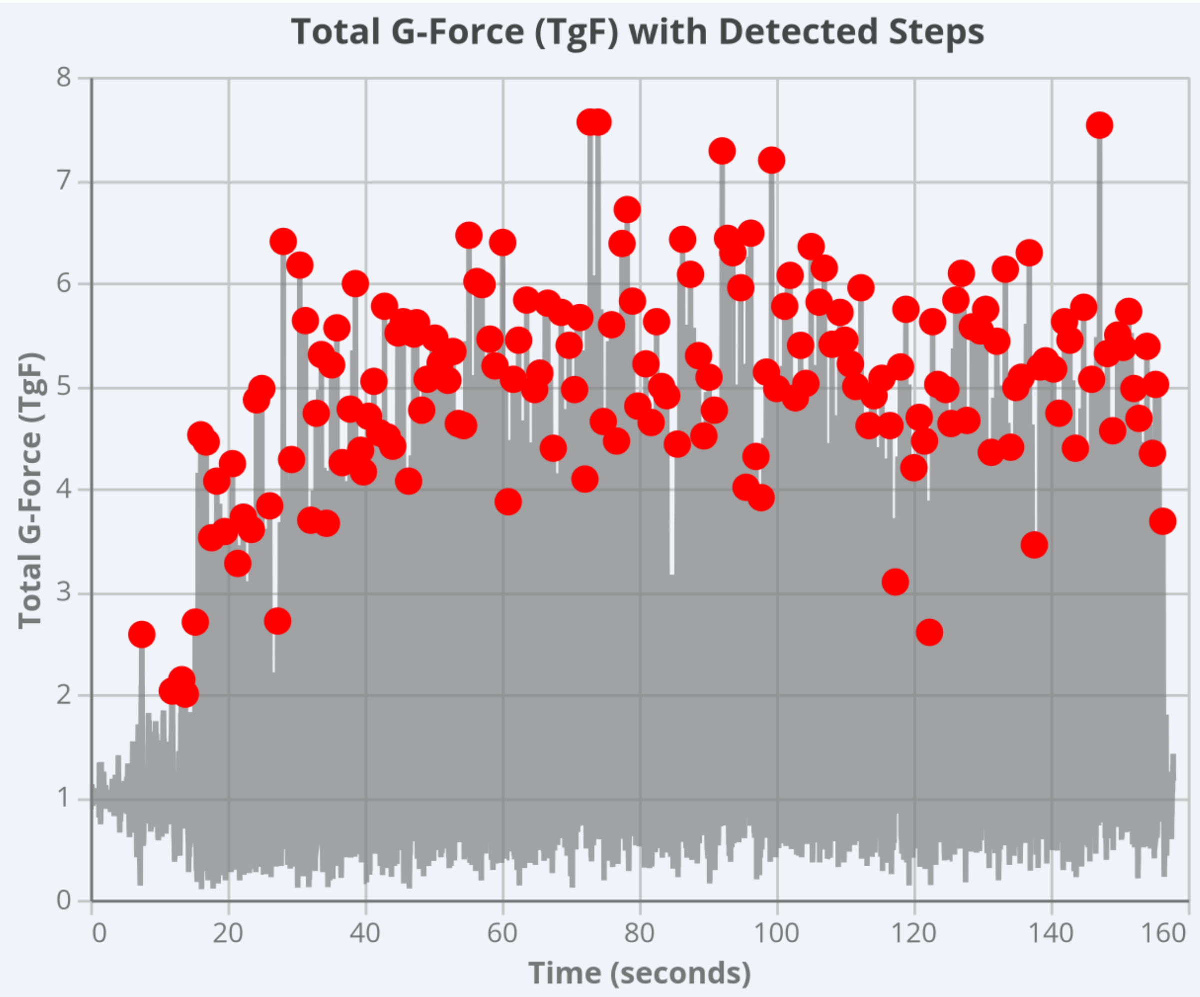
The aim of this assignment is to evaluate the accuracy of acceleration measurements obtained using smartphone sensors and to assess their effectiveness in collecting meaningful data for basic engineering analysis.

1.Motion Measurement Using Physics Toolbox Sensor Suite

Acceleration data was recorded while jogging on the inner track at IIT Jodhpur using the Physics Toolbox Sensor Suite. The total acceleration magnitude (TgF) was analysed to identify steps.

Step Detection and Gait Cycle Interpretation

A total of **167 distinct peaks** were detected in the acceleration signal.



{Analysis of the excel data sheet from the Physics Suite App}

- **Gait Cycle:** One complete cycle from the landing of one foot to the next landing of the same foot. A single prominent peak in TgF corresponds to one gait cycle.
- **Individual Steps:** Each foot contact with the ground (left step + right step).

If the detected 167 peaks represent **167 full gait cycles**, then the total number of individual steps is:

$$\text{Total steps} = 167 \times 2 = 334 \text{ steps}$$

Total Distance covered = 400 m (1 full round of the Ground)

2.Comparison with Pedometer Application

Given

- **Pedometer steps (Pedometer App):** 357
- **Graph-based steps (Physics Toolbox):** 334
- **Actual distance (ground):** 400 m
- **Pedometer distance:** 398 m

Percentage Error Calculation

$$\%\text{Error} = \frac{|\,400 - 398\,|}{400} \times 100$$
$$= \frac{2}{400} \times 100$$
$$= 0.5\%$$

Percentage Error in Step Count

$$\%\text{Error} = \frac{|\,357 - 334\,|}{357} \times 100$$
$$= \frac{23}{357} \times 100 \approx 6.44\%$$

Comment on Accuracy

The results show good accuracy, with step count error around **6%** and gravitational acceleration error less than **1%**, indicating that smartphone accelerometers are suitable for engineering measurements.
The percentage error in distance measurement is **0.5%**, indicating good accuracy of the pedometer measurement.

3.Measurement of Acceleration Due to Gravity

Details and specifications and sensitivity, accuracy of the accelerometer in your mobile phone and location of IIT Jodhpur

- The experiment was performed using a **Nothing Phone (3a)** equipped with a MEMS accelerometer.
- The sensor has a typical sensitivity of **0.01 m/s²** and an accuracy of approximately **±0.1 m/s²**, which is adequate for measuring human motion and gravitational acceleration.
- **Latitude (ϕ)**: 26.47° N
- **Longitude**: 73.11° E

Formula Used (Somigliana Equation)

$$g(\phi) = 9.780327(1 + 0.0053024\sin^2\phi - 0.0000058\sin^2 2\phi)$$

Substituting $\phi = 26.47^\circ$

$$g_{\text{theoretical}} \approx 9.79\ \text{m/s}^2$$

Experimental Measurement Using Smartphone

Using the accelerometer sensor in the Physics Toolbox Sensor Suite, the magnitude of gravitational acceleration measured when the phone was stationary was:

$$g_{\text{measured}} = 0.99\ G$$

Converting to SI units:

$$g_{\text{measured}} = 0.99 \times 9.81 = 9.71\ \text{m/s}^2$$

Accuracy Calculation

$$\%\text{Error} = \frac{|\, g_{\text{theoretical}} - g_{\text{measured}} \,|}{g_{\text{theoretical}}} \times 100$$
$$= \frac{|\, 9.79 - 9.71 \,|}{9.79} \times 100$$
$$\approx 0.82\%$$

Result

The theoretical acceleration due to gravity at IIT Jodhpur is approximately **9.79 m/s²**. The value measured using the smartphone accelerometer is **9.71 m/s² (0.99 G)**, resulting in an error of approximately **0.82%**.

How accurate is your measurement with respect to the computed value ?

The experimentally measured value of acceleration due to gravity differs from the computed theoretical value by less than 1%. This small deviation is mainly due to sensor noise and calibration limitations, and it indicates that the smartphone accelerometer provides reasonably accurate measurements for basic engineering applications.